

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

ERROR ANALYSIS TASK

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COURSE NAME: Deep Learning
FACULTY: Dr. Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

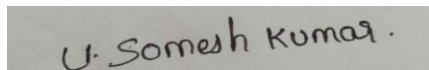
Assessment Tool Weightage: 30%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I U. Somesh Kumar (192425019) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS:

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

a) Identify the conceptual error in the likelihood formulation.

Answer:

Incorrect formulation:

$$L(\theta) = P(x_1) + P(x_2) + \dots + P(x_n)$$

MLE assumes independent observations, therefore likelihood must be the product of **probabilities**:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

The conceptual error is using sum instead of product.

b) Explain how this mistake affects parameter estimation.

Answer:

Using a sum does not represent the joint probability of all observations.

Correct MLE:

$$\hat{\theta} = \arg \max_{\theta} \prod_{i=1}^n P(x_i | \theta)$$

Wrong MLE:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n P(x_i | \theta)$$

This produces incorrect parameter estimates and reduces prediction accuracy.

c) Suggest the correct mathematical formulation and reasoning.

Answer:

Likelihood:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

MLE Estimate:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Because independent probabilities multiply to form the joint probability distribution.

d) How would using log-likelihood fix computational issues?

Answer:

Likelihood:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

MLE Estimate:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Because independent probabilities multiply to form the joint probability distribution.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

a) List possible implementation errors in the learning algorithm.

Answer:

Weight update rule may be wrong.

Correct rule:

$$w_{new} = w_{old} + \eta(y - \hat{y})x$$

Possible errors:

- Incorrect sign in update.
- Missing bias update.
- Wrong target labels.
- Incorrect activation threshold.

b) Analyze how incorrect learning rate or weight updates affect convergence.

Answer: .

Weight update:

$$\Delta w = \eta(y - \hat{y})x$$

If learning rate is too large:

$$\eta \uparrow$$

Weights oscillate and fail to converge.

If learning rate is too small:

$$\eta \downarrow$$

Training becomes extremely slow.

c) Identify dataset-related issues that may cause this problem.

Answer:

Although theoretically linearly separable, possible issues are:

- Label noise.
- Incorrect preprocessing.
- Duplicate conflicting samples.
- Data scaling problems.

These prevent convergence.

d) Propose modifications to ensure convergence.

Answer:

Normalize features:

$$x' = \frac{x - \mu}{\sigma}$$

Use proper learning rate:

$$0 < \eta < 1$$

Update bias correctly:

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

These modifications help guarantee convergence.

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

a) Identify possible mathematical or coding errors in gradient computation.

Answer:

Correct gradient:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

Possible errors:

- Wrong derivative.
- Missing chain rule.
- Wrong sign in gradient update.
- Incorrect indexing of weights.

b) Explain the role of activation functions in gradient flow.

Answer:

Sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Derivative:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Maximum value:

$$0.25$$

Small gradients slow learning.

ReLU:

$$f(x) = \max(0, x)$$

Derivative:

$$f'(x) = 1$$

Allows stronger gradient flow.

c) Analyse how vanishing or exploding gradients affect training.

Answer:

Vanishing Gradient:

$$(0.1)^5 = 0.00001$$

Gradient approaches zero.

Exploding Gradient:

$$(5)^5 = 3125$$

Gradient becomes excessively large.

Both prevent stable learning.

d) Suggest techniques to fix the issue (e.g., normalization, initialization).

Answer:

1. Xavier Initialization

$$Var(w) = \frac{1}{n}$$

2. He Initialization

$$Var(w) = \frac{2}{n}$$

3. Batch Normalization

$$x' = \frac{x - \mu}{\sigma}$$

4. Gradient Clipping

$$g = \min(g, threshold)$$

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

a) Explain why batch gradient descent is slow in large datasets.

Answer:

Gradient:

$$\nabla J(w) = \frac{1}{N} \sum_{i=1}^N \nabla L_i(w)$$

Every update requires processing all N samples.

Time complexity:

$$O(N)$$

per update.

Therefore, training is slow.

b) Analyse the cause of fluctuations in SGD.

Answer:

SGD update:

$$w_{t+1} = w_t - \eta \nabla L_i(w)$$

Only one sample is used.

Gradient estimate contains noise:

$$\nabla L_i(w) \neq \nabla J(w)$$

This causes fluctuations in loss.

c) Compare the error behaviour of Mini-Batch Gradient Descent.

Answer:

Mini-Batch Gradient:

$$\nabla J(w) = \frac{1}{m} \sum_{i=1}^m \nabla L_i(w)$$

where

$$m = 32,64,128$$

Advantages:

$$Variance_{MiniBatch} < Variance_{SGD}$$

and

$$Cost_{MiniBatch} < Cost_{BatchGD}$$

d) Recommend the best approach and justify with error trade-offs.

Answer:

Mini-Batch Gradient Descent.

Because:

$$BatchGD = Stable + Slow$$

$$SGD = Fast + Noisy$$

$$MiniBatch = Fast + Stable$$

Therefore Mini-Batch provides the best trade-off.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

a) Explain how high dimensionality increases model error.

Answer:

When dimensions increase:

$$d \gg n$$

Number of parameters increases significantly.

Generalization Error:

$$Error_{test} > Error_{train}$$

leading to overfitting.

b) Identify signs of overfitting in this scenario.

Answer:

Example:

Training Accuracy:

98%

Testing Accuracy:

70%

Gap:

$$98 - 70 = 28\%$$

Large gap indicates overfitting.

c) Analyse why distance-based learning fails in high dimensions.

Answer :

Distance:

$$D = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$$

As dimensions increase:

$$D_{max} \approx D_{min}$$

Distances become similar.

Nearest neighbours become difficult to distinguish.

d) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer:

PCA

Reduce dimensions:

$$10000 \rightarrow 100$$

L2 Regularization

$$J(w) = Loss + \lambda ||w||^2$$

Dropout

$$p = 0.5$$

Randomly disables neurons during training.

Result

- Reduced overfitting
- Better generalization
- Improved test accuracy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand